

SOC Home Lab

MITRE ATT&CK Detection Lab

*A Complete Guide Including Every Error, Every Fix,
and Every Command*

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Location: Lahore, Pakistan

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SIEM: Wazuh 4.7.5 | Target: Windows 7 | Attacker: Kali Linux

Techniques: T1078 | T1098 | T1484 | T1562.001 | T1059 | T1110

Detection Rate: 4 / 6 — 67%

This guide documents the complete journey of building a SOC home lab from scratch — including every error encountered, every fix applied, and every command used. Designed so any cybersecurity professional can replicate it.

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Project Overview

What and Why

What Is This Project?

This project documents the complete process of building a SOC home lab and using it to simulate real cyberattacks based on the MITRE ATT&CK framework. The goal was to practice detecting attacks the same way a real SOC Analyst would in a professional environment — including identifying and documenting cases where detection failed.

Why Build a SOC Home Lab?

Reason	Detail
Hands-on experience	Certifications teach theory. A lab teaches reality.
Real detection skills	Industry-standard tools like Wazuh used in production SOC environments.
Portfolio evidence	Concrete proof of skills for recruiters and hiring managers.
Troubleshooting practice	Core SOC analyst daily skill — problems will always occur.
Understanding attack behavior	How attacks actually appear in logs — not just theory.
Detection gap analysis	Identifying and documenting what a SIEM cannot see — equally important.

What Was Simulated?

Technique ID	Name	Method	Result
T1078	Valid Accounts	Created local attacker account and logged in	Detected ✓
T1098	Account Manipulation	Added account to Administrators group	Detected ✓
T1484	Policy Modification	Modified local group membership	Detected — Level 12 ✓
T1562.001	Impair Defenses	Stopped and restarted Wazuh agent service	Detected ✓
T1059	Command Execution	PowerShell reconnaissance commands	Not Detected ✗
T1110	Brute Force	Hydra SMB wordlist attack from Kali	Not Detected ✗

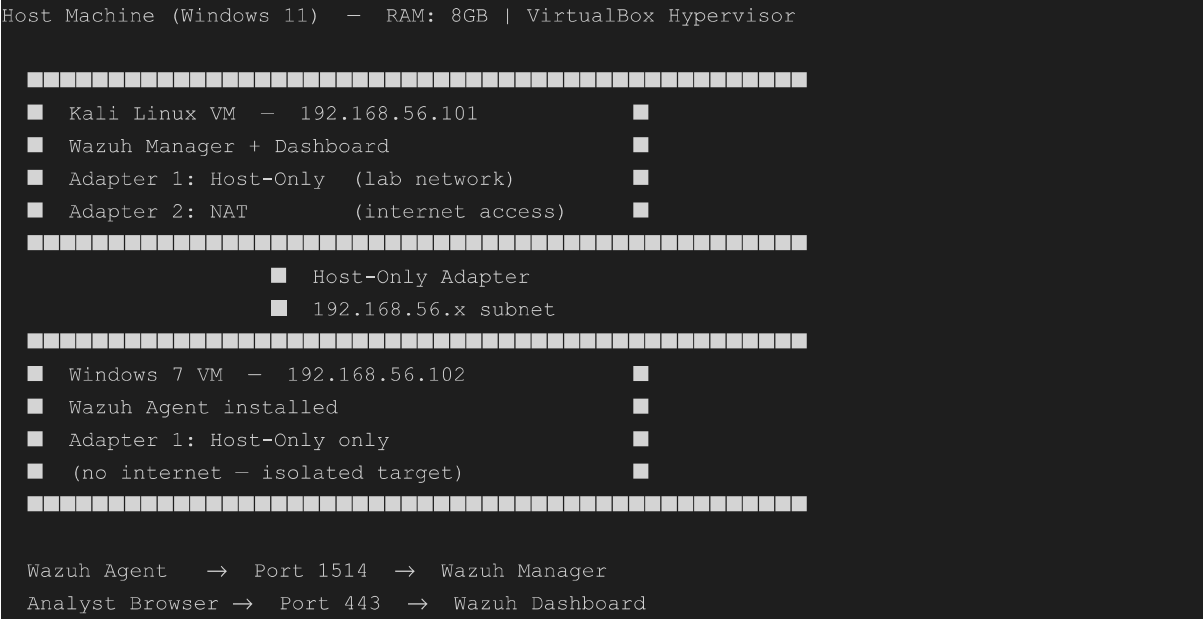
Chapter 2

Lab Architecture

Setup and Network Diagram

Component	Details	Purpose
Attacker Machine	Kali Linux 2025.3 (VirtualBox) IP: 192.168.56.101	Execute MITRE ATT&CK; techniques Run Wazuh SIEM manager
Victim Machine	Windows 7 (VirtualBox) IP: 192.168.56.102 Wazuh Agent installed	Target endpoint Generate security logs Simulate compromised machine
SIEM	Wazuh 4.7.5 Manager + Indexer + Dashboard Port: 443 (Dashboard)	Centralized log collection Rule-based alert correlation MITRE ATT&CK; mapping
Network	Host-Only Adapter Subnet: 192.168.56.0/24	Isolated VM-to-VM communication No internet exposure
Hypervisor	VirtualBox Host: Windows 11 (8GB RAM)	Virtual environment management Snapshot and restore capability

Network Diagram



KEY INSIGHT Kali Linux needs TWO network adapters — Adapter 1 (Host-Only) for lab communication with Windows 7, and Adapter 2 (NAT) for internet access to download tools. Windows 7 only needs Host-Only — internet is not required on the target machine.

Wazuh Installation

Setting Up SIEM on Kali Linux

What Is Wazuh?

Wazuh is a free, open-source SIEM (Security Information and Event Management) platform. It collects logs from endpoints, applies detection rules, maps alerts to MITRE ATT&CK, and provides a web dashboard for security event analysis.

Component	Function
wazuh-manager	Core engine — receives logs, applies rules, generates alerts
wazuh-indexer	OpenSearch-based storage — stores all security events
wazuh-dashboard	Web interface — visualize alerts and MITRE ATT&CK; mappings

Installation Command

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh && sudo bash wazuh-install.sh -a -i
```

Flag	Meaning
curl -sO	Download the install script silently
sudo bash wazuh-install.sh	Run the script as root
-a	All-in-one install (manager + indexer + dashboard)
-i	Ignore system compatibility check (required for Kali Linux)

Expected Installation Output

```
INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
INFO: --- Wazuh indexer ---
INFO: Wazuh indexer installation finished.
INFO: wazuh-indexer service started.
INFO: --- Wazuh server ---
INFO: Wazuh manager installation finished.
INFO: wazuh-manager service started.
INFO: --- Wazuh dashboard ---
INFO: Wazuh dashboard installation finished.
INFO: wazuh-dashboard service started.
INFO: --- Summary ---
INFO: You can access the web interface https://:443
User: admin
Password: [generated - SAVE THIS]
INFO: Installation finished.
```

KEY INSIGHT

Save the admin password that appears at the end of installation. If you lose it, you will need to reset it using the `wazuh-passwords-tool` script. Open the dashboard at <https://127.0.0.1> and accept the certificate warning — Wazuh uses a self-signed certificate.

The RAM Crisis

Wazuh Indexer Crash and Fix

The Problem

The host laptop had 8GB RAM. Running both Kali Linux VM and Windows 7 VM simultaneously, while also running Wazuh (which includes the resource-intensive OpenSearch indexer), caused the wazuh-indexer service to consume all available memory and crash repeatedly.

Error Observed:

```
systemctl status wazuh-indexer

Active: failed (Result: signal) since Mon 2026-04-27
Main PID: 764 (code=killed, signal=KILL)
Mem peak: 2.1G
```

The signal=KILL means the Linux kernel's OOM (Out of Memory) killer terminated the process because the system ran out of RAM. This is not a Wazuh bug — it is a resource constraint.

The Fix

Step	Action	Command
1	Shut down Windows 7 VM	VirtualBox → Windows 7 → Power Off
2	Start Wazuh indexer alone	sudo systemctl start wazuh-indexer
3	Wait for full initialization	sudo systemctl status wazuh-indexer
4	Start remaining Wazuh services	sudo systemctl start wazuh-manager wazuh-dashboard
5	Verify dashboard is accessible	Open https://127.0.0.1 in browser
6	Start Windows 7 VM	VirtualBox → Windows 7 → Start

KEY INSIGHT

This became a recurring workflow throughout the lab. Whenever Wazuh services needed to be restarted, Windows 7 was shut down first, Wazuh stabilized, then Windows 7 was restarted. In a production environment with proper hardware this would not be an issue.

Chapter 5

Network Configuration

VM-to-VM Communication Problem

The Problem

Initially both VMs were configured with NAT network adapters. NAT allows each VM to access the internet through the host machine, but VMs on NAT cannot communicate directly with each other. This meant Windows 7 could not reach the Wazuh manager on Kali, and the agent could not register or send logs.

```
# On Windows 7, trying to reach Kali:
ping 10.0.2.15
Request timed out.

# Browser on Windows 7:
https://10.0.2.15/agents-preview/
This page can't be displayed
```

Understanding the Solution

Adapter Type	Internet Access	VM-to-VM	Use Case
NAT	Yes	No	VM needs internet but isolation is fine
Host-Only	No	Yes	VMs need to talk to each other (lab environment)
Bridged	Yes	Yes	VM needs to be on same network as host

The Fix — Step by Step

Step 1 — Configure Kali with TWO adapters:

Adapter	Setting
Adapter 1	Host-Only Adapter
Adapter 2	NAT

Step 2 — Configure Windows 7 with ONE adapter:

Adapter	Setting
Adapter 1	Host-Only Adapter

Step 3 — Verify IPs after starting VMs:

```
# On Kali:
ip a | grep 'inet ' | grep -v 127
# Expected: inet 192.168.56.101/24 ... eth0

# On Windows 7:
ipconfig
```



```
# Expected: IPv4 Address: 192.168.56.102  
  
# Test connectivity from Windows 7:  
ping 192.168.56.101  
# Expected: Reply from 192.168.56.101
```

Agent Deployment

Transferring MSI Without Internet

The Problem

After switching Windows 7 to Host-Only adapter, it lost internet access. The standard Wazuh agent installation requires downloading an MSI file from the internet. With no internet on Windows 7, this was not possible directly.

The Solution — Python HTTP Server

Kali had two adapters — Host-Only (for lab network) and NAT (for internet). This meant Kali could download the MSI file from the internet AND serve it to Windows 7 over the Host-Only network using a simple Python HTTP server.

Step 1 — Download MSI on Kali:

```
wget https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.5-1.msi
```

Step 2 — Start Python HTTP server on Kali:

```
python3 -m http.server 8080  
# Output: Serving HTTP on 0.0.0.0 port 8080
```

Step 3 — Download from Windows 7 browser:

```
Open Internet Explorer on Windows 7:  
http://192.168.56.101:8080  
Click wazuh-agent-4.7.5-1.msi to download.
```

Step 4 — Install and configure the agent:

```
Double-click downloaded MSI → follow wizard  
Open 'Manage Agent' from Start menu  
Manager IP: 192.168.56.101 → Save → Manage > Start
```

Step 5 — Verify connection in Wazuh dashboard:

```
Open https://127.0.0.1  
Check: Total agents = 1, Active agents = 1
```

KEY INSIGHT

When you run `python3 -m http.server 8080`, Python creates a simple web server that serves all files in the current directory. Any device on the same network can browse to `http://[kali-ip]:8080` and download those files. This is a standard technique used in penetration testing and incident response for file transfer.

Challenges Reference

All Problems and Fixes

Challenge	Root Cause	Resolution
Wazuh indexer crashes (SIGKILL)	8GB host RAM insufficient for dual VMs + OpenSearch indexer simultaneously	Shut down Windows 7 VM before starting Wazuh; restarted after indexer stabilized
VM-to-VM communication fails	Both VMs on NAT — NAT provides internet but blocks VM-to-VM routing	Switched both VMs to Host-Only Adapter (192.168.56.x subnet)
Kali lost internet after Host-Only switch	Host-Only adapter does not route to internet	Added second adapter to Kali (Adapter 2: NAT) — Windows 7 kept Host-Only only
Wazuh agent MSI download fails on Windows 7	No internet access on isolated Windows 7 VM	Used Python HTTP server on Kali: <code>python3 -m http.server 8080</code>
Wazuh dashboard shows 'not ready yet'	Indexer not fully initialized — dashboard depends on indexer	Waited 3-5 minutes after indexer started before accessing dashboard
Copy-paste not working between host and VM	VirtualBox Guest Additions not installed	Installed Guest Additions via Devices menu; used manual typing for short commands
T1059 not detected despite correct setup	Windows 7 Event Channel does not forward process creation (Event ID 4688) reliably	Documented as detection gap — OS-level telemetry limitation, not Wazuh misconfiguration
T1110 brute force not detected	SMB failed logon events (4625) not consistently generated or forwarded	Documented as detection gap — requires threshold correlation rules and consistent logging
Wazuh security index corrupt	Admin password changed mid-session, breaking OpenSearch security index	Full reinstall: <code>sudo bash wazuh-install.sh -o -a -i</code>

T1078 — Valid Accounts

What Is T1078?

T1078 is a MITRE ATT&CK technique where an attacker uses legitimate credentials to log into a system. This is one of the most dangerous techniques because it blends in with normal user activity — the login itself looks identical to a legitimate user logging in.

After gaining initial access, attackers often create new local accounts to maintain persistence. By adding the account to the Administrators group, they ensure elevated access even if the original compromise is discovered and remediated.

Enable Audit Policy First

Before running any simulation, Windows audit policies must be enabled. Without this, zero security events are captured regardless of what happens on the system.

```
# Open PowerShell as Administrator on Windows 7:
auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
gpupdate /force

# Verify:
auditpol /get /subcategory:"Process Creation"
# Expected: Process Creation      Success and Failure
```

Simulation Steps

```
# Step 1 — Create attacker account
net user attacker Password123! /add

# Step 2 — Add to administrators group
net localgroup administrators attacker /add

# Step 3 — Login as attacker
runas /user:attacker cmd
# When prompted, enter: Password123!
```

✓ DETECTED

Field	Value
Technique	T1078 — Valid Accounts
Tactic	Defense Evasion, Persistence, Privilege Escalation, Initial Access
Event ID	4624

Field	Value
Event Type	Windows Logon Success
Target Username	attacker
Subject Username	Husnain
Logon Type	2 (Interactive)
Severity	Level 3 (Medium)
Rule IDs	60106, 60118
Agent	Husnain-PC (192.168.56.102)
AUDIT Result	AUDIT_SUCCESS

KEY INSIGHT

Wazuh correctly detected the logon event and mapped it to T1078. However, Event ID 4624 fires for every login — legitimate and malicious alike. Single-event detection is insufficient. Effective T1078 detection requires behavioral baselining: alert on first-time logons, off-hours access, or accounts that have never logged in before.

T1098 — Account Manipulation

Simulation and Detection

What Is T1098?

T1098 covers techniques where attackers manipulate accounts to maintain or expand access. This includes creating new accounts, modifying existing accounts, and changing group membership to escalate privileges. It falls under the Persistence tactic.

Simulation Steps

```
# Create second attacker account
net user attacker2 Husnain123 /add

# Add to administrators group
net localgroup administrators attacker2 /add
```

This simulation triggered TWO separate Wazuh alerts — one for account creation (T1098) and one for group membership change (T1484). This shows how a single action can map to multiple MITRE techniques.

✓ DETECTED

Field	Value
Technique	T1098 — Account Manipulation
Tactic	Persistence
Event Type	User Account Created + Group Membership Change
Target Account	attacker2
Severity	Level 8 (Medium-High)
Rule IDs	60109 (account created), 60110 (account changed)
AUDIT Result	AUDIT_SUCCESS

KEY INSIGHT

Privilege escalation through account manipulation is one of the most commonly overlooked activities in basic SIEM setups. In real environments, this technique generates noise because legitimate IT administrators perform the same actions routinely. Role-based access baselining and change management integration are critical.

T1484 — Domain Policy Modification

Simulation and Detection

What Is T1484?

T1484 covers modifications to group policies and security configurations. Adding accounts to privileged groups like Administrators triggers this technique. It falls under Defense Evasion and Privilege Escalation tactics.

How T1484 Was Triggered

T1484 was not simulated with a separate dedicated command — it was triggered automatically when attacker accounts were added to the Administrators group during T1078 and T1098 simulations. Wazuh's rule engine mapped the group membership change to T1484.

```
net localgroup administrators attacker /add  
net localgroup administrators attacker2 /add
```

✓ DETECTED

Field	Value
Technique	T1484 — Domain Policy Modification
Tactic	Defense Evasion, Privilege Escalation
Event ID	4732
Event Type	Administrators Group Changed
Target Group	Administrators (S-1-5-32-544)
Severity	Level 12 — HIGH (highest in this lab)
Rule ID	60154
Rule Desc	Administrators group changed
AUDIT Result	AUDIT_SUCCESS

KEY INSIGHT

Level 12 is the highest severity alert generated across all six techniques in this lab. In a production SOC environment, a Level 12 alert should trigger immediate analyst response and escalation — not just passive logging. An alert without a response playbook is just noise.

T1562.001 — Impair Defenses

Simulation and Detection

What Is T1562.001?

T1562.001 covers techniques where attackers disable or modify security tools to reduce detection coverage. Stopping the endpoint monitoring agent is one of the most direct methods — it creates a blind spot in SIEM visibility before executing noisier attack techniques.

Simulation Steps

```
# Open PowerShell as Administrator on Windows 7:

# Stop the Wazuh agent service
net stop WazuhSvc
# Expected: The Wazuh service was stopped successfully.

# Restart the agent
net start WazuhSvc
# Expected: The Wazuh service was started successfully.
```

When the agent stops, the Wazuh manager detects that the agent connection was lost. When the agent restarts, the manager detects the reconnection. Both events generate alerts and are mapped to T1562.001.

✓ DETECTED

Field	Value
Technique	T1562.001 — Impair Defenses: Disable or Modify Tools
Tactic	Defense Evasion
Event Type	Wazuh Agent Service Stop / Start
Alert 1	Wazuh agent stopped — Rule 506
Alert 2	Wazuh agent started — Rule 503
Severity	High
Detection Source	Wazuh manager internal monitoring

KEY INSIGHT

Wazuh successfully detected its own agent being stopped and restarted. Attackers cannot silently disable endpoint monitoring without being flagged — as long as the SIEM connection was active before tampering. However, if the attacker also clears Windows event logs simultaneously, this detection layer becomes much harder to rely on. Centralized log backup is essential.

T1059 — Command Execution

What Is T1059?

T1059 covers the use of command interpreters — PowerShell, CMD, scripts — to execute commands on a victim system. It is one of the most commonly used techniques in real cyberattacks. After gaining access, attackers use command-line tools to perform reconnaissance, establish persistence, and move laterally.

Simulation Steps

```
# Open PowerShell as Administrator on Windows 7:
whoami
# Output: husnain-pc\husnain

ipconfig
# Output: IP configuration details

net user
# Output: User accounts on the machine

systeminfo
# Output: OS version, patch level, hardware info
```

All commands executed successfully. Output was visible in PowerShell. From the attacker's perspective, the simulation was successful. But Wazuh generated zero alerts.

X NOT DETECTED

Investigation — This Was Not Accepted as a Simple Failure

Step	Action Taken	Result
1	Enabled Windows Audit Policy for Process Creation	Policy applied — confirmed with auditpol /get
2	Edited Wazuh ossec.conf — added Security event channel	Config saved and verified
3	Restarted Wazuh Manager and Wazuh Agent services	Services restarted successfully
4	Re-ran PowerShell simulation commands	Commands executed — still zero Wazuh alerts
5	Checked Windows Event Viewer manually	Event ID 4688 not consistently present
6	Researched Windows 7 Event Channel behavior	Root cause identified — OS limitation

Commands Used During Investigation:

```
# Step 1 - Enable audit policy:
auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable
gpupdate /force
auditpol /get /subcategory:"Process Creation"
# Output: Process Creation      Success and Failure

# Step 2 - Edit Wazuh config:
sudo nano /var/ossec/etc/ossec.conf
# Added before :
#
#     eventchannel
#     Security
#

# Step 3 - Restart services:
sudo systemctl restart wazuh-manager # On Kali
net stop WazuhSvc                    # On Windows 7
net start WazuhSvc
```

Root Cause Identified:

Windows 7 does not forward process creation events (Event ID 4688) through the Security event channel reliably. This is a known OS-level telemetry limitation. Windows 7 was released in 2009 — its event logging architecture predates modern SIEM requirements. Unlike Windows 10+, Windows 7 requires Sysmon or third-party agents to capture process creation at the level needed for T1059 detection.

Environment	T1059 Events	Why
Windows 10 + Splunk (parallel lab)	114 events	Modern OS — Event ID 4688 flows correctly through Security channel
Windows 7 + Wazuh (this lab)	0 events	Legacy OS — process creation not reliably forwarded via Security channel

KEY INSIGHT

The attack succeeded. The audit policy was correctly configured. The Wazuh rule was in place. Yet detection failed — because the log source itself was incomplete. This is exactly the kind of gap that exists in real SOC environments running legacy endpoints. Validate your log sources before trusting your detections.

T1110 — Brute Force

What Is T1110?

T1110 covers brute force attacks where an attacker repeatedly tries different passwords against a login service. Hydra automates this. SMB on port 445 is a frequent target because it exposes Windows authentication over the network.

Simulation Steps

```
# From Kali Linux terminal:

# Extract wordlist:
sudo gunzip /usr/share/wordlists/rockyou.txt.gz

# Run brute force attack:
hydra -l administrator -P /usr/share/wordlists/rockyou.txt 192.168.56.102 smb

# Hydra output:
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[DATA] max 1 task per 1 server, overall 1 task, 14344399 login tries
[DATA] attacking smb://192.168.56.102:445/
```

X NOT DETECTED

Root Cause Analysis:

Layer	Expected Behavior	Actual Behavior	Impact
Event Generation	Windows logs Event ID 4625 for each failed SMB logon	Inconsistent or no 4625 events from SMB	No raw data for Wazuh
Event Forwarding	Wazuh agent forwards 4625 events to manager	No events to forward	Wazuh sees nothing
Correlation	Wazuh aggregates failures into brute force alert	No events to correlate	No alert possible

What Effective Brute Force Detection Requires:

Requirement	Description
1. Consistent logging	Event ID 4625 must fire for every failed logon from every service
2. Threshold rule	Example: 5+ failures from same IP within 60 seconds = alert
3. Automated response	IP block, account lock, or escalation trigger after threshold

Requirement	Description
4. Source correlation	Multiple failures across accounts from same source = credential stuffing

KEY INSIGHT

Brute force detection is not about detecting a single failed login — it requires correlating multiple failures within a time window from the same source. Without that aggregation, each failed attempt looks like an isolated harmless event. The absence of this detection highlights a missing correlation layer that would need to be tuned in a production environment.

Findings Summary

All Results

Technique	Name	Result	Severity	Event ID	Key Finding
T1078	Valid Accounts	Detected ✓	Medium (L3)	4624	Logon events correlated — requires behavioral baselining
T1098	Account Manipulation	Detected ✓	Medium (L8)	4728	Group membership changes detected — noise in normal environments
T1484	Policy Modification	Detected ✓	HIGH (L12)	4732	Highest severity alert — Administrators group change flagged
T1562.001	Impair Defenses	Detected ✓	High	N/A	Agent service interruption detected in real time
T1059	Command Execution	Not Detected ✗	N/A	4688	Windows 7 telemetry gap — process creation not forwarded
T1110	Brute Force	Not Detected ✗	N/A	4625	SMB failed logons not aggregated — threshold correlation missing

Overall Detection Rate: 4 out of 6 techniques — 67%

Cross-Tool Comparison

Wazuh vs Splunk

This lab was run in parallel with a peer analyst (Wilfred Amimo) who built a similar lab using Splunk Enterprise + Windows 10 + KVM. Comparing results across different environments provides insight into how tool selection and OS version impact detection coverage.

Factor	This Lab	Parallel Lab (Wilfred Amimo)
SIEM	Wazuh 4.7.5	Splunk Enterprise
Target OS	Windows 7	Windows 10
Hypervisor	VirtualBox	KVM
Attacker	Kali Linux	Kali Linux (Host)
T1059 Events	0	114
T1110 Events	0	2
T1078 Events	8	8
Techniques Tested	6	3
Techniques Detected	4/6 (67%)	3/3 (100%)
Detection Gaps Documented	2 critical gaps with full root cause analysis	Not documented
OS Limitation Found	Yes — Windows 7 Event Channel gap	No (Windows 10 behaves correctly)
Key Advantage	Deeper gap analysis — more actionable for real SOC	Higher event volume on modern OS

KEY INSIGHT

Detection capability is not determined by tool selection alone. OS version, audit policy configuration, and log source integrity determine what a SIEM can and cannot see — regardless of how powerful the platform is. The 114 events for T1059 in the Splunk lab vs 0 in this lab is not a Wazuh limitation — it is a Windows 7 limitation. The same Wazuh configuration on Windows 10 would produce comparable results.

Lessons Learned

What This Lab Taught

1. Log source quality determines SIEM effectiveness.

Wazuh performed correctly — the failure was at the endpoint telemetry layer. A SIEM is only as good as the logs it receives. Before trusting any detection, validate the log source.

2. Default Windows logging is not SOC-ready.

Audit policies must be manually configured before a SIEM becomes useful. Without enabling process creation and logon auditing, entire attack categories go completely undetected.

3. Detection gaps must be actively investigated, not assumed.

T1059 was not accepted as 'just not working.' The audit policy was configured, Wazuh config was edited, services were restarted, and the root cause was traced to the OS event channel. That investigation process is what real SOC analysis looks like.

4. Context separates detections from noise.

T1078 and T1098 generated alerts — but the same events occur during legitimate administration. Real detection requires behavioral baselines, not just rule matching.

5. Missing detections are as valuable as successful ones.

The two undetected techniques revealed more about the monitoring environment than the four that were caught. Documenting blind spots is a core SOC responsibility — not optional.

6. OS version directly impacts detection coverage.

The parallel lab on Windows 10 detected T1059 with 114 events. Windows 7 generated zero. Tool selection and endpoint OS are inseparable from detection strategy.

7. Hardware constraints shape lab architecture.

Running Wazuh + two VMs on 8GB RAM required creative problem-solving. This constraint forced learning about resource management, service dependencies, and prioritization.

8. Troubleshooting is the real skill.

Every major error — RAM crashes, network failures, agent deployment problems — produced more learning than successful steps. In production SOC work, things always break.

Quick Reference

Wazuh Service Management

Action	Command
Start all Wazuh services	<code>sudo systemctl start wazuh-manager wazuh-indexer wazuh-dashboard</code>
Stop all Wazuh services	<code>sudo systemctl stop wazuh-manager wazuh-indexer wazuh-dashboard</code>
Check service status	<code>sudo systemctl status wazuh-manager</code>
Restart Wazuh manager	<code>sudo systemctl restart wazuh-manager</code>
Access Wazuh dashboard	Browser: <code>https://127.0.0.1</code>
View Wazuh logs	<code>sudo tail -f /var/ossec/logs/ossec.log</code>
Reinstall Wazuh (overwrite)	<code>sudo bash wazuh-install.sh -o -a -i</code>

Attack Simulation Commands

Technique	Command
T1078 — Valid Accounts	<code>net user attacker Password123! /add net localgroup administrators attacker /add runas /user:attacker cmd</code>
T1098 — Account Manipulation	<code>net user attacker2 Husnain123 /add net localgroup administrators attacker2 /add</code>
T1562.001 — Impair Defenses	<code>net stop WazuhSvc net start WazuhSvc</code>
T1059 — Command Execution	<code>whoami ipconfig net user systeminfo</code>
T1110 — Brute Force (from Kali)	<code>hydra -l administrator -P /usr/share/wordlists/rockyou.txt 192.168.56.102 smb</code>
Enable Audit Policy	<code>auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable gpupdate /force</code>

Wazuh Dashboard Filters

What to Find	Filter in Wazuh Dashboard
T1078 — Valid Account Logins	<code>rule.mitre.id: T1078 + data.win.system.eventID: 4624</code>
T1098 — Account Manipulation	<code>rule.mitre.id: T1098 + data.win.system.eventID: 4728</code>

What to Find	Filter in Wazuh Dashboard
T1484 — Group Change (High Sev)	rule.mitre.id: T1484 + rule.level: 12
T1562.001 — Agent Tampered	rule.mitre.id: T1562.001
All Husnain-PC alerts	agent.name: Husnain-PC
High severity only	rule.level: [10 TO 15]
All alerts last 24h	Time range: Last 24 hours

Troubleshooting Reference

Error Lookup Table

Error / Symptom	Cause	Fix
wazuh-indexer: failed (signal=KILL)	Out of memory — OOM killer terminated indexer	Shut down Windows 7 VM, restart wazuh-indexer alone, wait for stabilization
Dashboard: 'server is not ready yet'	Indexer not fully initialized	Wait 3-5 minutes after indexer starts; check: <code>systemctl status wazuh-indexer</code>
VMs cannot ping each other	Both VMs on NAT — blocks VM-to-VM communication	Switch both VMs to Host-Only Adapter in VirtualBox settings
Kali lost internet after Host-Only switch	Host-Only adapter has no internet routing	Add second adapter to Kali: Adapter 2 = NAT
Windows 7 cannot download Wazuh agent MSI	No internet on isolated Host-Only network	Download MSI on Kali, run: <code>python3 -m http.server 8080</code> , access from Windows 7 browser
Wazuh agent: 'Auth key not imported'	Agent installed but not registered with manager	In Agent Manager: enter Manager IP, click Save, then Manage > Start
T1059 — zero alerts despite correct setup	Windows 7 Event Channel limitation — 4688 not forwarded	Document as detection gap — upgrade to Windows 10+ or deploy Sysmon
T1110 — no brute force alerts	SMB 4625 events not consistently generated or aggregated	Enable threshold correlation rules; verify 4625 logging is active
Copy-paste not working in VM	VirtualBox Guest Additions not installed	Devices menu > Insert Guest Additions CD > install inside VM > reboot
Wazuh reinstall fails with 'already installed'	Previous installation still present	Use overwrite flag: <code>sudo bash wazuh-install.sh -o -a -i</code>
Wazuh security index corrupt	Admin password changed mid-session, breaking OpenSearch security index	Full reinstall with overwrite: <code>sudo bash wazuh-install.sh -o -a -i</code>

***The real SOC skill is not generating alerts.
It is knowing exactly why alerts are missing —
and documenting it with enough detail that someone else
can understand, replicate, and improve on your findings.***

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